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1
2 #include<Wire.h>
3 #include<SPI.h>
4 #include<LiquidCrystal_I2C.h>
5 #include<EEPROM.h>
6 LiquidCrystal_I2C lcd(0x27, 16, 2);
7 #include<Keypad.h>
8 #include<DS3231.h>
9 #include<LiquidCrystal.h>
10 DS3231 rtc(SDA,SCL) ;
11 Time t;
12 #define buzzerPin A0
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Fig. 3: Snippet of source code

TABLE 2 4×4 MATRIX KEYPAD CONNECTIONS	
4×4 Keypad Pin	Arduino Uno Pin
Row pin 0	D9
Row pin 1	D8
Row pin 2	D7
Row pin 3	D6
Col pin 0	D5
Col pin 1	D4
Col pin 2	D3
Col pin 3	D2

TABLE 3 DS3231 CONNECTIONS WITH UNO	
DS3231 Pins	Arduino Uno Pin
VCC	5V
GND	GND
SDA	A4
SCL	A5

TABLE 4 16×2 LCD CONNECTIONS WITH ARDUINO UNO	
12C LCD Pins	Arduino Uno Pin
VCC	5V
GND	GND
SDA	SDA
SCL	SCL



Fig. 4: Inside the prototype box

TABLE 5 RELAY CONNECTIONS WITH ARDUINO UNO	
Relay Pins	Arduino Pins
VCC	5V
GND	GND
IN	D10

TABLE 6 BUZZER WITH UNO	
Buzzer Pin	Arduino Uno
Buzzer positive	A0
Buzzer negative	GND

matches the real-time clock, the bell rings for a predetermined duration, which is set to approximately ten seconds in this implementation.

Software

This system requires libraries for the DS3231 RTC and I2C-based LCD, which need to be installed in Arduino IDE. To install these libraries:

Open Arduino IDE.

Navigate to sketch→Include library→Manage libraries.

Search for and install the following libraries:

DS3231

LiquidCrystal_I2C

Keypad

After installing the required libraries, download the source code from *electronicsforu.com*. Upload the code by selecting:

Board: Arduino Uno

COM Port: The port where the Arduino is connected

If all libraries (keypad, LCD display, and DS3231 RTC) are correctly installed, the code will upload successfully. Fig. 3 shows a snippet of the source code.

Next, the loop function is created to update the time from the RTC module, accept keypad inputs for setting bell timings and dates, and display the same on the LCD display. All settings are saved in EEPROM, ensuring they remain intact even after power is turned off.

Construction and setup

After uploading the source code into the Arduino Uno, assemble the circuit using Tables 2, 3, 4, 5, and 6 and the circuit diagram shown in Fig. 4. Enclose the entire setup in a small plastic box. Refer to Fig. 4 before assembling the prototype for a better understanding of the final build.

Power on the assembled unit by connecting the 9V adaptor to the Arduino Uno. The default day, date, and time will be displayed (see Fig. 5). Press '*' on the keypad to modify the settings.

Two options will appear:

Option 1: Change day, date, and time

Option 2: Set bell timings (see Fig. 6). If you want to change, press the '*' key using the keypad.

Press '1' to select 'SET YEAR'.

Use the keypad to enter the year, then press '#' (see Fig. 7).